Manual

Master terminal with DS-100 interfacing

TSW-MTS 8.1

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Contents

1 Master Terminal with DS-100 Interfacing - Overview

Possible fields of application

Along with DS-100 terminals and based on a shop floor PC, the master terminal function with DS-100 interfacing enables the decentralization of the automatic quantity input and the manual input of machine statuses.

Implementation notes

The function package is used if you:

- wish to implement decentralized quantity input based on a master terminal
- wish to implement a user-friendly option to manually enter the machine status directly at the machine

Integration

The function package requires an operative shop floor PC with installed data acquisition software (AIP or CTWIN).

Functions

Configurable application and communication software to operate a maximum of 16 DS-100 at PCs or shop floor terminals (Windows terminals or IPCs of other manufacturers) used as master terminal:

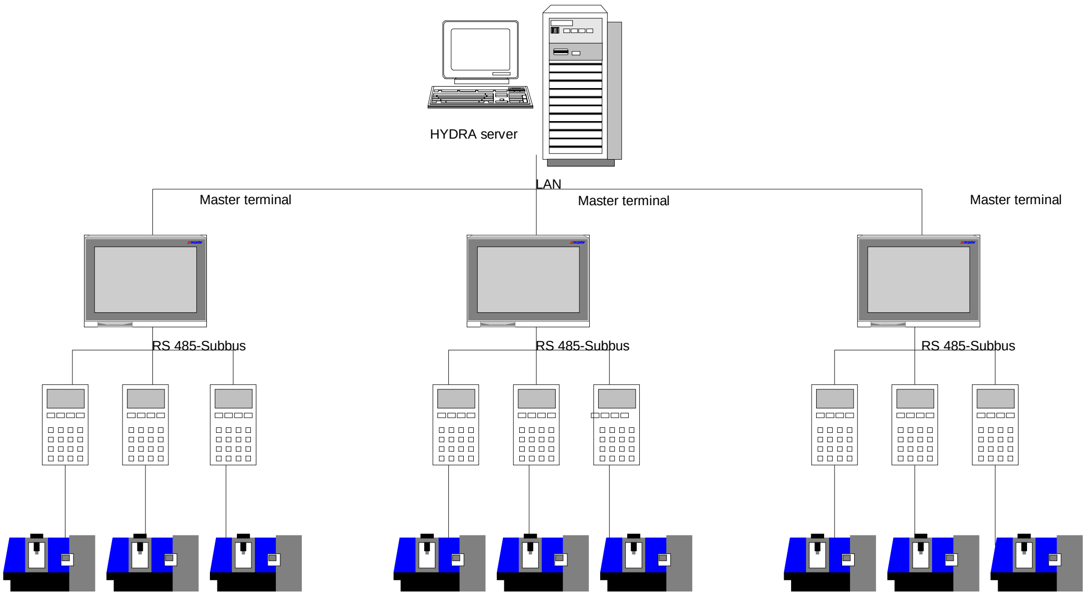
- Terminal dialogs to log on, interrupt and finish orders at a central place. Take-over of quantities and machine statuses recorded in a decentralized manner using DS-100.
- Current overview on the order progress (quantities) and machine statuses.

2 Master Terminal with DS-100

This document is based on the manuals dealing with the HYDRA standard Windows terminals (AIP or CTWIN). Consequently, this document only describes the special features if used as master terminal as well as the operation of DS-100.

The HYDRA standard Windows terminal can be used as master terminal to trigger the devices "DS-100" by MPDV as well as "MT3" by IBS. This results in a data acquisition system with the following structure:

Diagram
Master Terminal and DS-100 Subbus



However, it is not possible to use both device types at one master terminal.

3 Important information

A master terminal can trigger up to 16 devices of the type "DS-100" or "MT3".

3.1 DS-100 - Overview

- One DS-100 is exactly assigned to one machine.

- It provides the following interfaces for machine communication:
  - 2 counter inputs
  - 1 relay
  - 1 digital input

- DS-100 provides the following functions:
  - Import of machine signals (quantities/cycles, malfunction signals)
  - Input of downtime reasons
  - Operation of the production lock

- Staff as well as operations cannot be logged on to DS-100 terminals. These functions need to be performed at the master terminal.

- Due to its hardware properties, the DS-100 does not provide for an exact determination of actual cycles. This can only be achieved by using CT-MSS (stored program control, "machine interface"). Please contact MPDV Project Management for further information.

3.2 MT3 - Overview

- An MT3 is exactly assigned to one machine. Quantities are recorded using the machine interface (MSS).

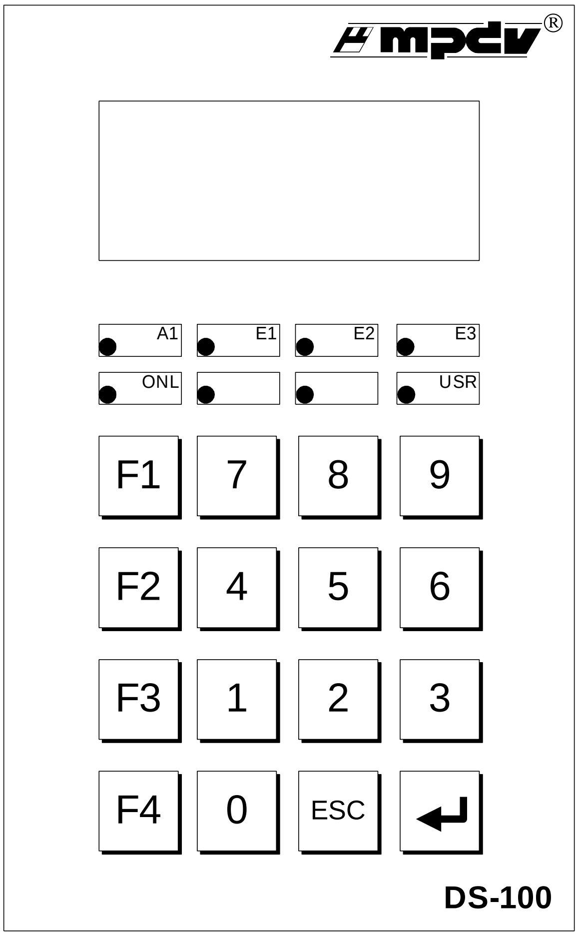
- **Please note:** The "production" status must have a number > 20 for MT3 devices.

4 Operation

4.1 Master terminal

The functions of the HYDRA standard Windows terminal are complemented by the "service display" (see below).

4.2 Layout of DS-100



Display

LED displays

Keypad

4.2.1 Functions of LED displays

A1	The display flashes if the internal relay "machine lock" is active. The configuration is performed at the HYDRA client (*ADE/MDE menu: Master data → Machine configuration → Machines/workplaces*; tab *MDE configuration → Inputs/outputs*; Option *machine lock*
E1	The display flashes if "counter 1" counts a pulse. "Master data→Machine/WP→MDE configuration--> yield counter=1" or "Master data→Machine/WP→MDE configuration --> scrap counter=1"

E2	The display flashes if "counter 2" counts a pulse. Configuration: like 'E1', but set counter to "2".
E3	The display flashes if the digital input of the terminal is set. This input has to be defined with "number 1" in the configuration of the HYDRA client.
LED1 / ONL	The display flashes shortly if the master terminal operates this device while polling.
LED2	The display flashes if the production lock is enabled for the machine.
LED3	Not assigned
LED4 / USR	The display flashes if the device waits for user input. e.g.: a machine status needs to be assigned

4.2.2 Keypad

F1	Enables input of a new machine status. The input fields are emptied.
F2	Not assigned
F3	Not assigned
F4	Switches the production lock.
0..9	Downtime: Input of the status number for status assignment Production: No input possible
↵	Confirmation of the input → Status number is transferred to the master terminal
ESC	Function of a reset button (the figure entered at last is deleted)

4.2.3 Display

The display includes four lines with 16 characters each.

Description:

Status	**Line1**	**Line2**	**Line3**	**Line4**
Production / OP logged on	Order no. + OP no.	Yield	Scrap	"Production“
Production without OP	(free)	Yield	Scrap	"Production:__“
Downtime / OP logged on	Order no. + OP no.	"Downtime“	Current status	"Status:___“
Downtime / no OP logged on	(free)	"Downtime“	Current status	"Status:___“
Downtime / production lock active	Order no. + OP no. / (free)	"Downtime“	Current status+ "PSP“	"Status:___“
Downtime / wrong status input	Order no. + OP no. / (free)	"Downtime“	Not available	"Status:___“
Shift break	(free)	"Downtime“	"No shift"	(free)

Examples:

Production / no order logged on

```

Yield:  1234
Scrap:    33
Production
```

Production / Order logged on

```
123456789012345
Yield:  1234
Scrap:    33
Production
```

Downtime / no status assigned

```

Downtime
Not assigned
Status:___
```

Downtime / no status assigned / production lock active

```

Downtime
Not assigned PSP
Status:___
```

Downtime / status assigned

```

Downtime
Setup       PSP
Status:___
```

Please note for displaying the order/operation:

In case more than one operation is still logged on after an interruption/logoff, the "next" operation that is displayed on the DS-100 is incidental.

4.2.4 Configuration of the DS-100 device address

- Pull out and plug in again the 9-pole bus connector.
- Click the F1 button within two seconds --> the current device address will be displayed in the hex format top left.

- By clicking the buttons F1 and F2, the displayed address will be increased or reduced.
- The displayed address is taken over by clicking "¿ ".
- The bus connector has to be pulled out again briefly to save the new device address.

Please note for the device address:

- The terminal is configured in hexadecimal notation

- The HYDRA client is configured in decimal notation The below table compares figures in decimal and hexadecimal notation to make this clear:

Dec. :	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hex. :	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Dec. :	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Hex. :	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Dec. :	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
Hex. :	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Dec. :	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
Hex. :	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F

e.g. **Dec.** 15 ==> **Hex.** 0F;   **Dec.** 16 ==> **Hex.** 10.

5 Configuration

5.1 Configuration on the HYDRA client

5.1.1 Terminal configuration

ADE/MDE: Master data > Terminal configuration > Terminals

- Tab *General*
  - The terminals "CT760" as well as "CT830" can be used as master terminals. Consequently, the terminal type has to be set to "760" or "830".
  - The operation mode of the terminal has to be set to "MDE terminal".
- Tab *MDE*
  - The *Master terminal* option has to be set.

5.1.2 MDE configuration of the machine

ADE/MDE: Master data > Machine/workplace configuration > Machines/workplaces

- Tab *MDE configuration > Inputs/outputs*
  - Provided that the option "master terminal" was set in the terminal configuration, "external connection" shows the option DS-100 or MT3 as well as the device address. The device address is to be entered as decimal number.
  - **DS-100 only:** The relay for the machine lock is set at DS-100 if the value "1" is set in the field "machine lock". As DS-100 has only one channel, 1 must always be entered there. The relay is then set for all statuses assigned to “machine lock” as well as for the “not assigned” status.

Please note: It is not possible to operate DS-100 and MT3 devices at one terminal.

5.2 Entries in the configuration file of the master terminal

The following entries are relevant for the master terminal function, in addition to the INI file settings of the Windows shop floor terminal described in the documentation dealing with the standard Windows terminal:

Entry	Comment
Section [system]	
`UpdateTime=1`	MDE is requested every 5 seconds by default. As polling of the assigned devices is triggered by this request, the request cycle should be reduced to one second by setting this value for the operation mode "master terminal". (by default: 5)
Section [comports]	
`Com1=MASTER`	The serial interfaces used for triggering the devices, have to be entered as MASTER.

6 Service display at the master terminal

The service screen can be opened by entering the shortcut "Ctrl+Alt+L“ in the basic screen of the master terminal:

ID	DS/MT	Maschine	V	NET	Reset	Status(h)	E1-Z1	E2-Z2	E3	pol	FKT	Eingabe
00	DS	6001	/	J	08.04.99 13:23	4048	65336	65535	0	1	D	3
01	DS	6002	/	J		4044	65535	65535	0	1		
02	DS	6003	/	J		4040	65535	65535	0	1		
03	DS	6004	/	J		4042	65535	65535	0	1		
04	DS	6005	/	J		4040	65535	65535	0	1		
05	DS	6006	/	J		4042	65535	65535	0	1		
06	DS	6007	/	J		4042	65535	65535	0	1		
07	DS	6008	/	J		4042	65535	65535	0	1		
08	DS	6009	/	J		4040	65535	65535	0	1		
09	DS	6010	/	J		4042	65535	65535	0	1		
10	DS	6011	/	J		4040	65535	65535	0	1		
11	DS	6012	/	J		4040	65535	65535	0	1		
12	DS	6013	\	J		4044	65535	65535	0	1		45
13	DS	6014	/	J		4042	65535	65535	0	1		
14	DS	6015	/	J		4042	65535	65535	0	1	D-	
15	DS	6016	/	J	08.04.99 13:05	4042	65535	65535	0	1		
			/									

Communication with the connected devices can be tracked in this view. Not all fields are filled out with MT3 devices.

Field	Description
ID	Device name
DS/MT	Device type (DS-DS-100 / MT-MT3)
Machine	Machine number of the assigned machine
V	Monitoring of the communication while polling: The device that is currently being operated shows a backslash character "\“.
NET	Connection status (J: device responds / T: timeout) (not with MT3)
Reset	Start date and time are entered here if a device has been restarted since starting of the master terminal. (not with MT3)
Status(h)	Internal DS-100 device status (not with MT3)
E1-Z1	Meter reading of counter 1 (counted down from 65535) (not with MT3)
E2-Z2	Meter reading of counter 1 (not with MT3)
E3	Status of the digital input (1-set) (not with MT3)

Field	Description
Pol	Number of devices started during the last run (the device is started several times if it responds to a status input). (Not with MT3)
FKT	Shows the pressed function keys
Input	Shows the status value entered in the bottom line (not with MT3)